DO NOW · AO1 REVISION

Energy resources

Energy
Paper 1 · Combined and Separate
29 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 14 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Name the three main uses for energy resources.**	Transport, electricity generation and heating	☐
2 ★	**What is a renewable energy resource?**	One that is being (or can be) replenished as it is used.	☐
3	**Name the nine main energy resources.**	Fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, geothermal, tides, the Sun and water waves	☐
4 ★	**Name the three types of fossil fuels.**	Coal, oil and natural gas	☐
5 ★	**Identify the two main non-renewable energy resources.**	Fossil fuels and nuclear fuel	☐
6	**Name two energy resources which involve burning or combustion.**	Fossil fuels and biofuel	☐
7 ★	**State what reliable means for an energy resource.**	It can supply energy whenever it is needed.	☐
8 ★	**Explain why solar power is not reliable in every country.**	The amount of sunlight varies with the weather and the time of day.	☐
9	**Explain why wind power is not reliable in every country.**	The wind speed varies, and sometimes there is no wind.	☐
10 ★	**Explain why fossil fuels are considered reliable.**	They can be burned at any time, in varying amounts, to match demand.	☐
11	**Explain why hydroelectricity is considered reliable.**	The flow of water through a dam can be adjusted depending on demand.	☐
12	**Identify the energy resource that contributes most to climate change.**	Fossil fuels	☐

© 2026 PanPhy Labs. All rights reserved.

	QUESTION	ANSWER	✓
13	**Other than climate change, name two environmental problems that energy resources can cause.**	Any two from: habitat destruction, air pollution, radioactive waste, visual or noise pollution	☐
14	**Name two energy resources that can fuel internal combustion engines.**	Fossil fuels and biofuel	☐
15	**Name the energy resources that can be used for electricity generation.**	Fossil fuels, nuclear fuel, biofuel, wind, hydroelectricity, solar, geothermal, tidal and wave	☐
16	**Name the energy resources that can be used for heating.**	Fossil fuels, biofuel, geothermal and solar	☐
17	**State four types of consideration that affect the choice of energy resource.**	Environmental, political, social and economic	☐
18 ★	**What is a non-renewable energy resource?**	One that is not replenished as it is used, so it will run out.	☐
19	**Explain why nuclear fuel is a non-renewable energy resource.**	The supply of uranium and plutonium is finite and is not replenished.	☐
20 ★	**Give one advantage of nuclear power over burning fossil fuels.**	It produces no carbon dioxide when generating electricity.	☐
21 ★	**Give one disadvantage of nuclear power.**	It produces radioactive waste that stays hazardous for a very long time.	☐
22 ★	**Name the gas from burning fossil fuels that contributes most to climate change.**	Carbon dioxide	☐
23 ★	**Other than climate change, state one environmental problem caused by burning fossil fuels.**	Acid rain (caused by sulfur dioxide)	☐
24 ★	**State one environmental impact of a hydroelectric dam or tidal barrage.**	It floods land or changes habitats, harming wildlife.	☐
25 ★	**Explain why tidal power is more reliable than wind power.**	The tides are predictable; the wind is not.	☐
26	**Explain why scientists cannot always deal with the environmental issues they identify.**	Because of political, social, ethical or economic considerations.	☐

© 2026 PanPhy Labs. All rights reserved.

	QUESTION	ANSWER	✓
27 ★	**Explain why biofuel can be described as carbon neutral.**	The plants absorb as much carbon dioxide growing as is released when the fuel burns.	☐
28	**Give one disadvantage of geothermal power.**	It is only possible in a few places.	☐
29	**Compare the costs of renewable energy resources with fossil fuels.**	Expensive to set up, but no fuel cost and cheap to run.	☐

© 2026 PanPhy Labs. All rights reserved.